STAT 5630 Statistical Machine Learning

Cross-City Urban Heat Hotspot Screening:

Within-City Learning and City-Held-Out Transfer

Final Report

Collaborators

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Main Text

1. Background Information

Extreme urban heat is a public-health, infrastructure, and planning problem because thermal exposure is not distributed evenly across a city. At fine spatial scales, differences in pavement, roof surfaces, vegetation, water proximity, terrain, building density, and local land cover can produce large contrasts in surface heat across neighborhoods that are close together. A citywide average therefore hides the places where heat-mitigation investments, additional monitoring, or planning attention may be most useful. For a statistical learning project, the practical question is not only whether a city contains hot areas, but whether publicly available spatial variables can help rank local cells for screening when direct thermal labels are limited.

Much urban heat mapping is local and descriptive: it shows where hotter surfaces occur within one city or estimates associations between LST and surface descriptors in familiar places. The statistical question in this report is broader: whether basic environmental and built-environment features can identify urban heat hotspots across a multi-city dataset, and how the answer changes when evaluation moves from held-out cells inside represented cities to whole-city holdout transfer. That framing makes both the standardized multi-city dataset and the validation design contrast central to the contribution rather than just pipeline details.

Remote sensing is central to this problem because satellites can observe spatial patterns that are difficult to measure with ground stations alone. Thermal infrared sensors measure emitted radiation from the land surface and can be converted into land surface temperature, or LST. LST is not the same as the two-meter air temperature reported in a weather forecast: it describes how warm the observed surface would feel to the touch, so a roof, road, tree canopy, irrigated field, or bare soil patch may all behave differently. NASA's land-surface-temperature documentation emphasizes that land heats and cools differently from the air, and ECOSTRESS documentation describes a thermal mission that measures surface temperature at fine spatial detail from the International Space Station. That distinction is important for this report. The target is based on surface temperature, so the results should be interpreted as surface hotspot screening rather than direct human heat-exposure measurement.

Urban thermal remote sensing has a long history, but the literature also shows why this project needs a careful validation design. Voogt and Oke (2003) review thermal remote sensing of urban climates and note that many urban thermal studies historically emphasized qualitative maps of thermal patterns and simple correlations with surface descriptors. Later Landsat-based work gave more quantitative evidence for specific predictors. Weng, Lu, and Schubring (2004) studied the relationship between land surface temperature and vegetation abundance in Indianapolis and treated NDVI as a standard indicator of vegetation abundance in urban heat-island analysis. Yuan and Bauer (2007) compared NDVI and impervious surface as indicators of surface urban heat-island effects in the Twin Cities and found that percent impervious surface had a strong relationship with LST across seasons, while NDVI remained useful but more seasonally variable. Stewart and Oke (2012) also show why urban thermal behavior depends on local surface form, land cover, and urban structure rather

than climate alone. Together, this work motivates the feature set used here: vegetation, imperviousness, land cover, terrain, and water proximity are not arbitrary columns, but physically plausible surface descriptors.

[Insert related-work extension here: briefly connect additional urban heat, spatial transfer, or spatial machine-learning literature to the report's motivation, emphasizing why the report compares within-city hotspot screening with city-held-out transfer rather than treating either design as sufficient by itself.]

Vegetation matters because evapotranspiration and shading can lower surface temperatures, while low vegetation often leaves more exposed built or bare surfaces. Impervious surfaces matter because roads, roofs, sidewalks, and parking areas store and reradiate heat and reduce evaporative cooling. Land-cover class provides a compact description of surface type, distinguishing developed, forested, grass, wetland, and other covers. Water proximity matters because water bodies and riparian corridors can be associated with local cooling or with land-cover transitions. Terrain is also relevant: elevation, slope context, and valley or coastal settings can influence radiation, drainage, and local thermal patterns. Heat vulnerability also depends on people, resources, and exposure conditions, not surface temperature alone, as shown in neighborhood-scale heat-stress work such as Harlan et al. (2006). This report uses elevation directly and keeps distance-to-water as a simple hydrologic-context variable, while avoiding a larger feature expansion in the headline model so the first benchmark stays interpretable.

The data-access workflow also follows established remote-sensing practice. AppEEARS, the Application for Extracting and Exploring Analysis Ready Samples, is a NASA Earthdata/LP DAAC tool for spatial, temporal, and layer subsetting of remote-sensing products. In this project it supports reproducible area requests for MODIS/Terra NDVI and ECOSTRESS LST over city study regions. ECOSTRESS provides atmospherically corrected LST and emissivity products, while AppEEARS helps reduce those products to the city polygons and seasonal windows used for analysis. This matters for reproducibility because the report does not rely on manually downloaded images; it uses consistent acquisition and preprocessing rules tied to the same city definitions used for modeling.

The validation design follows from that gap. Even when prediction models are used, ordinary random row or cell splits can be too easy for spatial data because nearby cells and cells from the same city can appear in both training and testing. Roberts et al. (2017) argue that ignoring spatial, temporal, or hierarchical structure in cross-validation can underestimate predictive error, and Meyer et al. (2018) show that target-oriented validation can reveal much weaker performance than random folds when models must predict beyond familiar locations. This project applies the same principle at the city scale while also retaining the easier same-city screening question: within-city held-out evaluation asks whether the feature set can recover local hotspot structure when city examples are available, while city-held-out transfer asks whether those relationships carry to cities whose labels were completely unseen during training.

This project therefore contributes a course-scale benchmark with three linked pieces: a standardized multi-city dataset, an explicit comparison between within-city held-out screening and city-held-out transfer, and a transparent model comparison using simple baselines, lo-

gistic regression, and random forest. The completed dataset contains 71,394,894 rows and 17 columns, with one row representing one analytic 30 m grid cell in one of 30 selected U.S. cities. The cities are balanced across three broad climate groups: 10 hot-arid, 10 hot-humid, and 10 mild-cool cities. The city-held-out benchmark remains the stricter transfer result, but it is interpreted alongside the within-city result rather than as a separate project. Figure 1 shows the selected city locations, and Table 2 summarizes the final audited dataset by climate group.

The dataset combines public geospatial and remote-sensing sources that describe both the urban surface and the thermal outcome. 2020 Census urban-area geometry defines each study region. Annual NLCD 2021 land-cover and impervious-surface products provide categorical land cover and built-intensity measures. USGS 3DEP elevation supplies terrain information, while NHDPlus HR hydrography supports a distance-to-water feature. MODIS/Terra MOD13A1.061 NDVI, acquired through AppEEARS for May 1-August 31, 2023, summarizes warm-season vegetation. ECOSTRESS ECO_L2T_LSTE.002 land surface temperature, also acquired through AppEEARS for the same May-August 2023 window, provides the thermal measurement used to define the hotspot label. Table 1 lists these source layers and the final variables constructed from them.

The modeling setup deliberately excludes the thermal variables from the predictive feature set. ECOSTRESS LST is used to define the outcome, not to predict it. The first-pass headline models use non-thermal, transferable predictors: impervious percentage, land-cover class, elevation, distance to water, median May-August NDVI, and broad climate group. This feature contract makes the prediction task intentionally constrained, avoids leaking the answer, and keeps the model closer to a screening workflow that could be applied before collecting city-specific hotspot labels.

2. Research Questions

The primary research question is:

Can basic environmental and built-environment features identify urban heat hotspots across a multi-city dataset, and how does performance change when evaluation moves from within-city held-out cells to whole-city held-out transfer?

This question has two explicit parts. The within-city screening question asks whether, when a city is represented during model development, the model can identify held-out hotspot cells from that same city. The city-held-out transfer question asks whether, when a whole city is excluded from training, the model can rank that unseen city's cells by hotspot risk. The unit of analysis is a 30 m grid cell, but the stricter transfer evaluation uses city as the grouping unit: a held-out city must remain fully unseen while model preprocessing, imputation, scaling, encoding, and tuning are learned from other cities. The target population for inference is 30 m grid cells inside buffered Census urban-area study regions for the selected U.S. cities, with cautious extension to similar urban areas.

The project also asks four secondary questions. First, do non-thermal geospatial predictors contain useful signal for the locally hottest urban grid cells? Second, does a nonlinear random

forest improve hotspot screening compared with logistic regression and simple baselines? Third, does model performance vary systematically by city or climate group? Fourth, does success under within-city evaluation predict success under city-held-out transfer?

The outcome is `hotspot_10pct`, a city-relative binary label. A positive cell is one of the hottest 10% of valid eligible cells within its own city, based on median May-August ECOSTRESS LST after final filtering. This is not an absolute national heat threshold. A positive cell in Boston and a positive cell in Phoenix both represent the upper tail of their local city distributions, even though their absolute temperatures may differ. That choice makes the target appropriate for relative within-city screening across a cross-climate city set, where a practical planning question is often which local cells should be prioritized for closer inspection.

3. Dataset Construction

The dataset construction workflow starts with city selection and study-region definition. The project uses 30 U.S. cities, balanced across three broad climate groups: hot-arid, hot-humid, and mild-cool. Cities were selected to provide broad geographic and climate variation while remaining feasible for a standardized data-acquisition and modeling pipeline. The set should therefore be treated as a purposive benchmark panel rather than a statistically representative sample of U.S. urban areas. Performance differences across the selected cities are informative for transfer diagnostics, but they should not be interpreted as estimates of national average performance. Ten cities were assigned to each group to create a transfer test spanning dry western metros, humid southern and eastern metros, and cooler or milder metros. The climate groups are intentionally coarse design labels. They are not meant to replace formal climatology or local meteorology, but they make the transfer question more interpretable by asking whether a model trained across one mixture of cities behaves differently across broad regional climate contexts. This balance also prevents the benchmark from becoming only a large-city or only a hot-desert exercise.

For each selected city, the study region begins with the 2020 Census urban area containing the city center. That urban-area polygon is used as the core geometry, and the analysis also preserves it separately from the buffered study area. The default study area then applies a 2 km buffer around the core urban area. The buffer matters because heat patterns, water adjacency, and built/natural land-cover transitions do not stop exactly at administrative or Census boundaries. It is a fixed design choice rather than an optimized parameter, so it can include some fringe land differently across cities. Preserving the unbuffered core geometry keeps the original urban-area definition available for comparing core-city and buffered-area filters.

Each city receives a master 30 m grid built in a local UTM coordinate reference system. This is a practical geospatial choice. A local projected CRS makes distances and cell areas meaningful within a city, which is necessary for 30 m cell construction and for the distance-to-water feature. The grid is the analytic backbone of the project: every final row is one grid cell, and every raster or vector source is converted into a value for that same cell. Without

a master grid, differences in source resolution, projection, and pixel alignment could create hidden inconsistencies across variables.

The phrase "30 m dataset" therefore refers to the analytic grid and row unit, not to the native resolution of every source variable. NLCD is naturally close to the grid scale, but MODIS/Terra NDVI is coarser, ECOSTRESS has its own thermal-pixel and overpass structure, and vector-derived quantities such as distance to water are summarized to cell centroids or cell geometry. The resulting table is valuable because it gives every city a common modeling unit, but it should not be interpreted as 71.4 million independent 30 m native-resolution measurements of every phenomenon.

The source layers are then prepared against this grid. NLCD 2021 land cover supplies a categorical surface class for each cell, and NLCD 2021 imperviousness supplies a continuous percent-impervious value. USGS 3DEP elevation is aligned to the grid to provide terrain context. NHDPlus HR hydrography is handled as vector information: water features are clipped to the city study area and converted to a distance from each grid cell to the nearest selected hydro feature. MODIS/Terra NDVI and ECOSTRESS LST are acquired through AppEEARS using the city area of interest and summarized for May-August 2023. The project uses median May-August NDVI as the vegetation predictor and median May-August ECOSTRESS LST as the thermal outcome ingredient.

The May-August seasonal summary is a compromise between heat relevance and data robustness. It focuses the target and vegetation feature on the warm season, when surface heat is most relevant to the urban heat problem, while aggregating over multiple observations rather than depending on one satellite pass. ECOSTRESS also has irregular overpass timing because it is mounted on the International Space Station, so the analysis records `n_valid_ecostress_passes` as a quality-support field. Cells with fewer than three valid ECOSTRESS observations are removed when LST is available. This filter reduces the chance that a cell is labeled from an unstable or nearly missing thermal summary, but pass count alone does not make all cities observationally identical. Overpass time, clouds, seasonality within May-August, and weather conditions can still vary across cities and remain part of the remote-sensing limitation.

The final assembly step merges the per-city feature tables into a single modeling-ready dataset, with CSV also written as a compatibility output. Open-water cells are dropped where `land_cover_class == 11`, because open water is a different thermal surface and would complicate the urban land-cell screening target. After the water and ECOSTRESS pass-count filters, the hotspot label is recomputed within each city. This recomputation is essential: the positive class should represent the hottest 10% of the eligible modeling population in that city, not the top decile of a larger pre-filtered set that included cells no longer eligible for modeling.

The target, `hotspot_10pct`, is therefore city-specific. A cell is positive if its median May-August LST falls in the top 10% among valid eligible cells in its own city. This design supports cross-climate comparison because it asks about local hotspot screening rather than one absolute national LST threshold. A Phoenix hotspot and a Boston hotspot may have different absolute surface temperatures, but both represent locally high surface temperature within their urban context. The top-decile target is useful for screening because it turns

each city into a ranked local prioritization problem. The tradeoff is that the target should not be interpreted as an absolute heat-risk standard or as a direct planning action without local validation.

The final dataset contains the city identifier and name, broad climate group, cell identifier, cell centroid longitude and latitude, six headline predictors, ECOSTRESS-derived LST and pass count, the `hotspot_10pct` label, and three neighborhood-context variables constructed for later model extensions. The main benchmark keeps the six-feature specification fixed and does not use those neighborhood variables in the logistic-versus-random-forest comparison. This separation is important because it lets richer-feature experiments be reported as extensions rather than silent changes to the main benchmark.

The completed audit confirms 30 cities, 71,394,894 rows, and 7,139,588 hotspot-positive cells. Because `hotspot_10pct` is recomputed within city, overall prevalence is approximately 10% by construction, and each climate group has nearly the same hotspot prevalence despite large differences in total row count. The feature-missingness summary is favorable for the headline predictors: imperviousness, land cover, distance to water, and climate group have no missing values in the final table; elevation is missing for only 3,426 cells, and median May-August NDVI is missing for 99,625 cells, about 0.14% of the dataset. The row-count distribution is uneven across cities because study-area extents differ substantially; Appendix Figure A1 is included as support for that point.

The large row count should be interpreted with the 30 m analytic-grid caveat above. Nearby cells share land cover, repeated coarser-source summaries, sensor conditions, spatial context, and thermal structure, so the effective independent information is smaller than the raw row count. The city-held-out design addresses one major source of leakage across cities, but it does not erase within-city spatial dependence.

Figure 2 summarizes the end-to-end dataset construction workflow. The key modeling implication is that the final table is not an ad hoc spreadsheet: it starts from public data sources, aligns every layer to a 30 m city grid, applies explicit row filters, and creates a target suitable for testing generalization to unseen cities.

4. Model and Method

Given that construction, the statistical learning task is to estimate, for each grid cell, the probability that the cell belongs to the hottest 10% of valid eligible cells in its own city. The response variable is `hotspot_10pct`, and the headline models use only the six non-thermal predictors in the main benchmark: `impervious_pct`, `land_cover_class`, `elevation_m`, `dist_to_water_m`, `ndvi_median_may_aug`, and `climate_group`.

Several columns are intentionally excluded from the predictive feature set. The target itself, `hotspot_10pct`, cannot be used as an input. The thermal variables `lst_median_may_aug` and `n_valid_ecostress_passes` are also excluded because LST defines the label and ECOSTRESS pass count is a data-quality support variable rather than a stable urban-form predictor. Cell identifiers, city identifiers, city names, and centroid coordinates are excluded

because they could let a model memorize location-specific patterns instead of learning a portable relationship between urban surface characteristics and hotspot status.

The report separates two validation designs because they answer different scientific questions. The first is within-city held-out evaluation: cities are represented during model development, and models are evaluated on held-out cells from those same cities. The retained within-city summary uses a verified 70/30 held-out structure and reports class-1 hotspot precision, recall, and F1 for logistic regression and random forest. This design measures same-city hotspot screening or interpolation. It is valuable because it tests whether the feature set contains local hotspot signal when the model has access to examples from the same urban systems it is asked to screen.

[Insert within-city held-out design details and results here: describe the verified 70/30 within-city evaluation, report logistic-versus-random-forest precision, recall, and F1, and explain that this setting measures same-city hotspot screening rather than unseen-city transfer.]

The second design is city-held-out transfer, shown schematically in Figure 3. The 30 cities are partitioned into five deterministic outer folds, with six complete cities held out in each fold and the remaining 24 cities used for training. Every city is held out exactly once. This design is stricter than a random cell-level split because all rows from a test city are unseen during model fitting. For each outer fold, all preprocessing, imputation, scaling, categorical encoding, feature construction inside the model pipeline, and hyperparameter tuning are fit using training-city rows only. The held-out cities are touched only after the full training pipeline has been selected and fit.

This train-city-only rule is the core leakage control for the transfer benchmark. Numeric imputation values, scaling parameters, categorical encodings, and tuned hyperparameters are all learned without looking at the held-out cities. The rule also applies to model selection: inner cross-validation for tuning is performed only inside the training-city portion of each outer fold. As a result, a held-out city's labels cannot influence preprocessing choices, feature encodings, or tuning decisions. The final held-out prediction table is therefore a transfer test at the city level, not a same-city interpolation exercise. The city-held-out benchmark remains the stricter headline result for transfer, while the within-city result provides the comparison needed to interpret how validation design changes apparent performance.

The report compares simple transfer baselines, logistic regression, and random forest. The simple baselines provide sanity checks for whether a learned model improves beyond very limited rules. The baseline set includes a no-skill prevalence reference, a global-mean baseline, a land-cover-only baseline, an impervious-only baseline, and a climate-only baseline. These baselines are useful because they show how much performance can be obtained from single-feature or prevalence-style transfer rules before fitting a richer model.

The logistic regression model is the main linear benchmark. Missing-value imputation, numeric scaling, one-hot encoding of categorical features, and the classifier are fit together inside each training fold. The classifier uses the `saga` solver, which supports regularized logistic regression over the six-feature set. In course terms, logistic regression models the log-odds of hotspot status as an additive function of the predictors. It is useful here because it is relatively interpretable and because a strong linear model is a fair reference point for

asking whether nonlinear structure adds value.

The random forest model is the main nonlinear benchmark. It uses the same training-only preprocessing rule. Numeric predictors are imputed, categorical predictors are encoded inside the training fold, and the forest is tuned over tree count, tree depth, feature subsampling, and minimum leaf size. A random forest averages predictions from many decision trees, allowing threshold effects and interactions among variables such as land cover, vegetation, imperviousness, and climate group. The model is therefore a natural test of whether nonlinear relationships improve city-held-out screening under the same six-feature contract.

The two validation settings use related but not identical metrics, so the report compares patterns and implications rather than treating all numbers as one leaderboard. Within-city held-out results are summarized with thresholded hotspot precision, recall, and F1. City-held-out transfer uses precision-recall area under the curve, or PR AUC, as the primary ranking metric. PR AUC summarizes how well a model ranks true hotspots above non-hotspots across possible score thresholds. It is more informative than raw accuracy in this project because the positive class is intentionally about 10% of cells. A trivial model that predicts "not hotspot" for every cell would be highly accurate but useless for finding hotspots. PR AUC focuses instead on the tradeoff between precision, the share of predicted positives that are true positives, and recall, the share of true positives recovered.

The report also uses mean city PR AUC and recall at the top 10% predicted risk. Pooled PR AUC weights cities roughly by their sampled held-out row counts, so larger cities can influence the aggregate more strongly. Mean city PR AUC first computes PR AUC for each held-out city and then averages across cities, giving each city equal interpretive weight. Recall at top 10% predicted risk is a screening-oriented metric: it asks what fraction of true hotspots would be recovered if a city inspected only the cells the model ranked in the highest-risk decile. Because the target itself is also a top-decile label, this metric is easy to interpret as top-decile retrieval.

Sampling and benchmark scope are important for interpreting the numbers. The benchmark results are sampled all-fold runs, not exhaustive model fitting and scoring over all 71,394,894 rows. The main comparison uses 5,000 rows sampled per city with target-rate stratification: 500 positives and 4,500 negatives per city, using random state 42. The same sampled city preload is used for training rows and held-out scoring rows, so each outer fold trains on 120,000 sampled rows from 24 cities and tests on 30,000 sampled rows from six held-out cities. The matched 5k comparison is the headline logistic-versus-random-forest comparison because both models share the same sample cap and fold design. A 20,000 rows-per-city logistic run is included only as higher-sample linear context. This sampling choice is a computational caveat, not a change in the grouped evaluation design: cities are still held out as complete units, and training-only preprocessing and tuning are still required.

The sampled design is also an inferential choice. It gives each city the same benchmark row count, which keeps train and test sizes comparable across held-out cities instead of letting the largest study areas dominate every model comparison. Target-rate stratification preserves the 10% hotspot prevalence in each sampled city, so PR AUC and top-decile recall are evaluated under the same class balance used to define the target. Because every city contributes the same number of positives and negatives, the retained benchmark is best

interpreted as a controlled comparison of model ranking behavior across cities, not as full-map operational performance. What it does not preserve exactly is full spatial density, all within-city clustering, or the exact full-population distribution of land cover, imperviousness, vegetation, and terrain. The retained benchmark should therefore be read as a sampled transfer-screening comparison, not exhaustive full-city scoring. Repeated samples or full-city scoring would be needed to quantify sampling variability.

5. Analysis, Conclusion and Discussion

The empirical story begins with the easier within-city held-out setting. In the verified 70/30 within-city evaluation summarized in Figure 4, random forest clearly outperforms logistic regression on class-1 hotspot precision, recall, and F1. Across the 30 cities, random forest reaches mean hotspot precision 0.7310, recall 0.3433, and F1 0.4480, compared with logistic regression means of 0.3887, 0.0727, and 0.1083. This result suggests that the six-feature contract contains real local hotspot signal and that nonlinear or interaction-like structure helps when local city examples are available during model development.

[Insert within-city held-out narrative detail here: add the full write-up of the verified 70/30 within-city design, including how cells were split, how thresholded hotspot classifications were produced, and how the precision, recall, and F1 results should be interpreted as same-city screening.]

The city-held-out transfer benchmark gives a weaker and closer result. The retained models rank hotspots above chance in held-out cities, but gains over imperviousness and land-cover baselines are small and concentrate mainly in selected hot-arid cities. As shown in Table 3 and Figure 6, the random-forest 5k model reaches pooled PR AUC 0.1486, above the 0.1000 prevalence reference and above the 0.1353 land-cover-only baseline. The logistic SAGA 5k model reaches 0.1421. These values are not large in absolute terms, which is expected for a whole-city transfer task with a 10% positive class, but they are consistently above a no-skill ranking reference on pooled PR AUC.

The cleanest transfer model comparison is the matched 5,000 rows-per-city logistic-versus-random-forest result. On pooled PR AUC, random forest improves from 0.1421 to 0.1486. On recall at the top 10% predicted risk, random forest improves from 0.1647 to 0.1961. At the same time, logistic regression remains slightly stronger on mean city PR AUC, with 0.1803 for logistic compared with 0.1781 for random forest. The 20,000 rows-per-city logistic run reaches pooled PR AUC 0.1457 and recall at top 10% of 0.1709, which provides useful context but is not the headline comparison because it uses a larger sample than the 5k random-forest run.

The prevalence context is important. Because the hotspot label is defined as the top 10% of eligible cells within each city, a low-signal ranking model has little room to look useful by accuracy alone. PR AUC is therefore a more relevant transfer summary than overall correct classification. A PR AUC modestly above 0.10 indicates that the model is ranking true hotspots above non-hotspots better than a prevalence-level reference, but it does not imply reliable cell-by-cell classification. On recall at the top 10% predicted risk, random forest reaches 0.1961, but the impervious-only baseline already reaches 0.1858. In practical

screening terms, much of the transferable retrieval signal in this six-feature contract is already captured by simple built-intensity information.

The main result is therefore not one universal model ranking. Within-city held-out evaluation makes random forest look clearly stronger, while city-held-out transfer is weaker, closer between models, and more heterogeneous. The side-by-side comparison in Figure 4 should be read as a validation-design result: the same broad modeling ingredients look different when the model is allowed to learn from the same cities it is tested on versus when whole cities are withheld. Within-city success should not be treated as evidence of cross-city transfer.

Figure 5 makes that point at the city level. The city-level correlation between within-city random-forest hotspot F1 and city-held-out random-forest PR AUC is 0.08. The correlation between within-city random-forest hotspot recall and city-held-out random-forest recall at top 10% is 0.03. These weak relationships mean that cities that look comparatively learnable under within-city testing do not reliably become the cities where whole-city transfer works best. The result separates "the model can learn local hotspot structure" from "that structure transfers cleanly from other cities."

[Insert evaluation-design signal-shift analysis here: expand the Figure 5 interpretation by comparing city-level within-city random-forest F1/recall against city-held-out random-forest PR AUC/recall@top10, reporting the weak correlations from the presentation, and explaining that same-city learnability does not imply cross-city transferability.]

The fold-level transfer results in Table 5 show why point estimates should not be overread. Random forest improves PR AUC in folds 0, 3, and 4, but trails logistic regression in folds 1 and 2. Its recall gains are also fold-dependent: positive in folds 0, 3, and 4, negative in folds 1 and 2. Table 6 gives the city-level paired summary. Across 30 cities, the mean RF-minus-logistic PR AUC delta is -0.0023, the median delta is -0.0136, and logistic wins 21 city-level PR AUC comparisons while random forest wins 9. For recall at top 10%, the mean delta is +0.0106 but the median delta is -0.0150, again with logistic winning 21 cities. This is transfer heterogeneity rather than a uniform nonlinear improvement.

City and climate-group summaries explain where the random-forest transfer gains appear. Table 4 shows that random forest has positive mean gains in hot-arid cities: +0.0336 in PR AUC and +0.0762 in recall at top 10% relative to the matched logistic model. In contrast, hot-humid cities average -0.0123 in PR AUC and -0.0164 in recall, while mild-cool cities average -0.0281 in PR AUC and -0.0280 in recall. Figure 7 shows the city-level RF-minus-logistic deltas behind those climate summaries. The largest random-forest gains occur in hot-arid cities such as Las Vegas, Bakersfield, Tucson, and Fresno. Several hot-humid and mild-cool cities favor logistic regression, with large random-forest losses in places such as San Jose, Chicago, Portland, and Atlanta. Figure 8 adds the absolute random-forest city PR AUC values and is important for avoiding an overreading of Figure 7: high RF-minus-logistic gain does not necessarily mean a city is easiest to predict; it can also reflect weaker logistic performance. The climate interpretation should be read as a hypothesis generated by the benchmark rather than as an established causal explanation.

Figure 9 provides a spatial diagnostic view of one held-out city, Denver. Denver is retained

as a representative hot-arid held-out example from the map outputs. The figure shows predicted top-decile risk, true hotspot cells, and categorical error types for a random-forest fold in which Denver was excluded from training. The model captures some spatial structure, but false positives and false negatives remain spatially organized. That pattern suggests the transferred model learns partial built-environment signal while missing city-specific detail. The map is useful for interpreting the transfer task spatially, not for claiming deployment readiness.

Supplemental feature-importance evidence is shown in Appendix Figure A2 and should be treated cautiously. The random-forest permutation summaries and logistic coefficients are consistent with vegetation, imperviousness, land cover, elevation, and climate group carrying predictive signal. That does not prove that changing any one feature would cause a specific LST response. The predictors are correlated with broader urban form, land management, local climate, and sensor-observation conditions. Their value in this report is predictive and diagnostic, not causal.

These results support a plain conclusion. The project contributes a reproducible 30-city aligned dataset, an explicit comparison of within-city held-out screening and city-held-out transfer, and a transparent benchmark for comparing hotspot-screening models. Public non-thermal geospatial predictors carry real hotspot signal, but the strength and meaning of that signal depend strongly on the validation design. Random forest is clearly stronger than logistic regression under within-city held-out evaluation, while the stricter city-held-out transfer benchmark is weaker, closer between models, and more heterogeneous. The contribution is therefore not simply that one model wins; it is that the project separates learnable same-city hotspot structure from cross-city transferability.

Validity can be organized into six parts. Leakage and internal validity are strongest for the city-held-out transfer benchmark because complete cities are held out and preprocessing and tuning are fit inside training cities. Sampling validity is bounded because the transfer benchmark uses 5,000 sampled rows per city rather than exhaustive all-row training and scoring; the sample preserves target prevalence and city balance, but not every spatial or feature-distribution detail. Spatial validity is improved by city holdout, but spatial dependence and clustered errors remain within held-out cities. Construct validity is limited by the LST-versus-exposure distinction and the 30 m analytic-grid caveat described earlier. External validity is limited because the 30 selected cities are a useful benchmark set, not a complete representation of all U.S. urban forms, climates, coastal settings, and topographies. Model-comparison validity is also bounded: within-city and transfer results use different evaluation designs and metrics, so their magnitudes should not be collapsed into one direct leaderboard.

Future work should extend both sides of the evaluation contrast. The most important next steps are to document the within-city split and thresholding procedure in full, score larger held-out samples or full held-out cities, add uncertainty summaries over cities, test whether neighborhood-context predictors improve transfer without overfitting city-specific structure, and compare the LST-based target with air-temperature, exposure, or vulnerability measures where such data are available.

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Tables and Figures

Tables and figures are organized here rather than interleaved with the main text. Appendix tables provide additional city, fold, and model-specification details.

Table 1. Data Sources and Constructed Variables

Source	Raw product / layer	Constructed final variable(s)	Spatial role	Used in headline model?
U.S. Census urban areas	2020 Census TIGERweb urban-area polygon containing each selected city center	Study-area and core-city geometry; 30 m city grid	Defines the city footprint, 2 km buffered study area, and grid alignment target	No
NLCD	Annual NLCD 2021 Collection 1 land-cover class raster	`land_cover_class`	Categorical built and natural surface-cover predictor; open-water filter where class 11 is present	Yes
NLCD	Annual NLCD 2021 Collection 1 impervious percentage raster	`impervious_pct`	Cell-level built-intensity predictor aligned to the 30 m grid	Yes
USGS 3DEP	3DEP 1 arc-second digital elevation model	`elevation_m`	Terrain predictor aligned to the 30 m grid	Yes
NHDPlus HR	NHDPlus High Resolution hydrography water features	`dist_to_water_m`	Distance-to-nearest-water predictor derived from clipped vector water features	Yes
MODIS/Terra via AppEEARS	MOD13A1.061 500 m 16-day NDVI observations, May 1-August 31, 2023	`ndvi_median_may_aug`	Summertime vegetation predictor summarized to each grid cell	Yes
ECOSTRESS via AppEEARS	ECO_L2T_LSTE.002 daytime land-surface-temperature observations, May 1-August 31, 2023	`lst_median_may_aug`; `n_valid_ecostress_passes`; `hotspot_10pct`	Thermal outcome source, LST quality support field, and within-city top-decile target	No

All constructed variables in Table 1 are ultimately summarized to the common 30 m analytic grid. As described in Section 3, this grid alignment should not be read as native 30 m measurement for every source layer.

Table 2. Final Dataset Summary by Climate Group

Climate group	City count	Total rows	Total hotspot positives	Hotspot prevalence	Min city rows	Median city rows	Max city rows	Median valid ECOSTRESS passes
Hot-arid	10	12,814,143	1,281,427	0.1000	382,964	1,100,156	3,199,440	30.0
Hot-humid	10	27,098,157	2,709,866	0.1000	700,063	1,788,622	7,081,699	21.5
Mild-cool	10	31,482,594	3,148,295	0.1000	817,627	2,889,018	6,722,963	33.0

Table 3. Main City-Held-Out Benchmark Metrics

Rows labeled 5,000 sampled use the same target-rate-stratified per-city sample and can be compared directly.

Model checkpoint	Rows per city	Pooled PR AUC	Mean city PR AUC	Recall at top 10%	Runtime (min)
No-skill / prevalence reference	5,000 sampled	0.1000	0.1000	0.1000	n/a
Global-mean baseline	5,000 sampled	0.0982	0.0997	0.0971	n/a
Climate-only baseline	5,000 sampled	0.0982	0.0997	0.0975	n/a
Impervious-only baseline	5,000 sampled	0.1351	0.1519	0.1858	n/a
Land-cover-only baseline	5,000 sampled	0.1353	0.1479	0.1672	n/a
Logistic SAGA 5k	5,000 sampled	0.1421	0.1803	0.1647	35.6
Logistic SAGA 20k context	20,000 sampled	0.1457	0.1796	0.1709	156.6
Random forest 5k	5,000 sampled	0.1486	0.1781	0.1961	97.2

Notes: the impervious-only baseline is the strongest simple baseline on recall, and the land-cover-only baseline is the strongest simple baseline on pooled PR AUC. The 5k logistic

model is the matched linear comparison for the 5k random forest, while the 20k logistic row is higher-sample linear context rather than the headline comparison. The global-mean and climate-only baselines are effectively constant or near-constant ranking rules, so small deviations around the 0.1000 prevalence reference reflect tie handling rather than meaningful below-reference skill.

Table 4. RF Minus Logistic Performance by Climate Group

Positive deltas mean the random forest performed better than the matched logistic model within that climate group.

Climate group	City count	RF PR AUC wins	Logistic PR AUC wins	Mean PR AUC delta	RF recall wins	Logistic recall wins	Mean recall-at-top-10% delta
Hot-arid	10	5	5	+0.0336	6	4	+0.0762
Hot-humid	10	2	8	-0.0123	2	8	-0.0164
Mild-cool	10	2	8	-0.0281	1	9	-0.0280

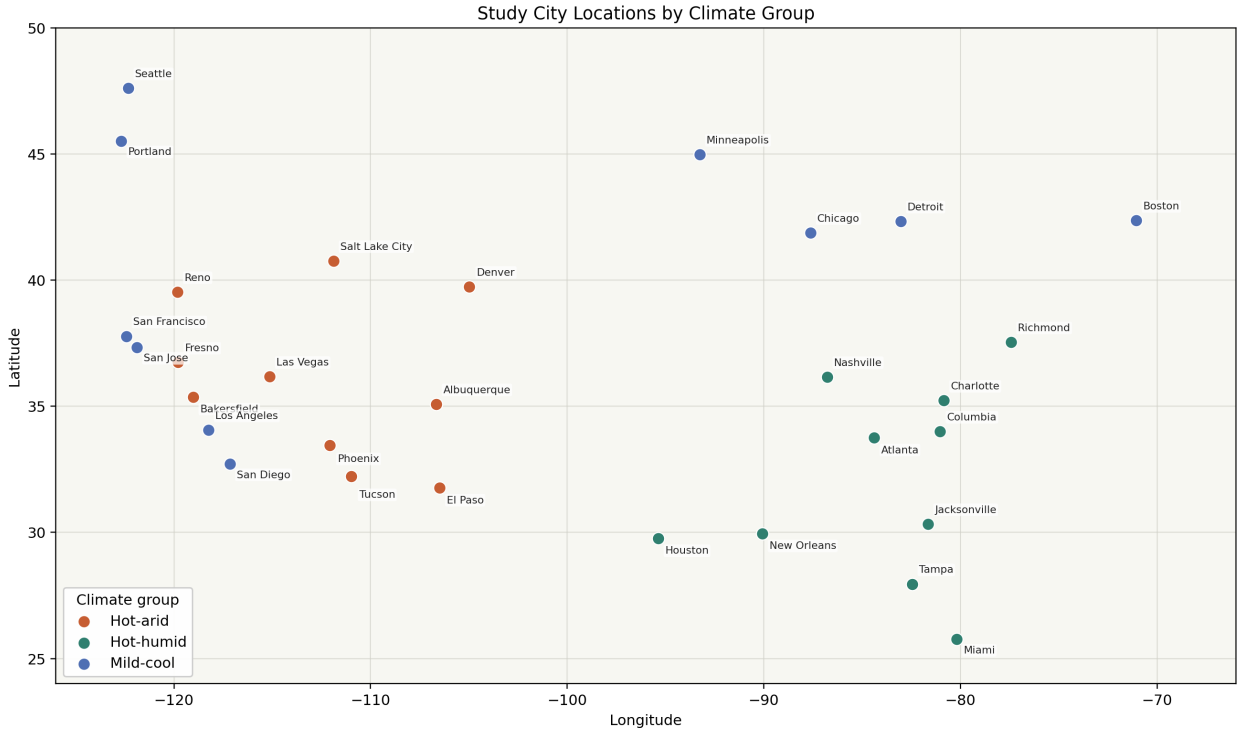
Table 5. Fold-Level RF Minus Logistic Comparison

Outer fold	Train rows	Test rows	Test positives	Test prevalence	Logistic PR AUC	RF PR AUC	RF minus logistic PR AUC	Logistic recall@top10	RF recall@top10	RF minus logistic recall@top10
0	120,000	30,000	3,000	0.1000	0.1610	0.1773	+0.0163	0.2170	0.2217	+0.0047
1	120,000	30,000	3,000	0.1000	0.2006	0.1598	-0.0408	0.2563	0.2087	-0.0477
2	120,000	30,000	3,000	0.1000	0.1436	0.1301	-0.0135	0.1777	0.1443	-0.0333
3	120,000	30,000	3,000	0.1000	0.1267	0.1606	+0.0340	0.1463	0.2133	+0.0670
4	120,000	30,000	3,000	0.1000	0.1471	0.1493	+0.0022	0.1640	0.2020	+0.0380

Table 6. City-Level Paired RF Minus Logistic Summary

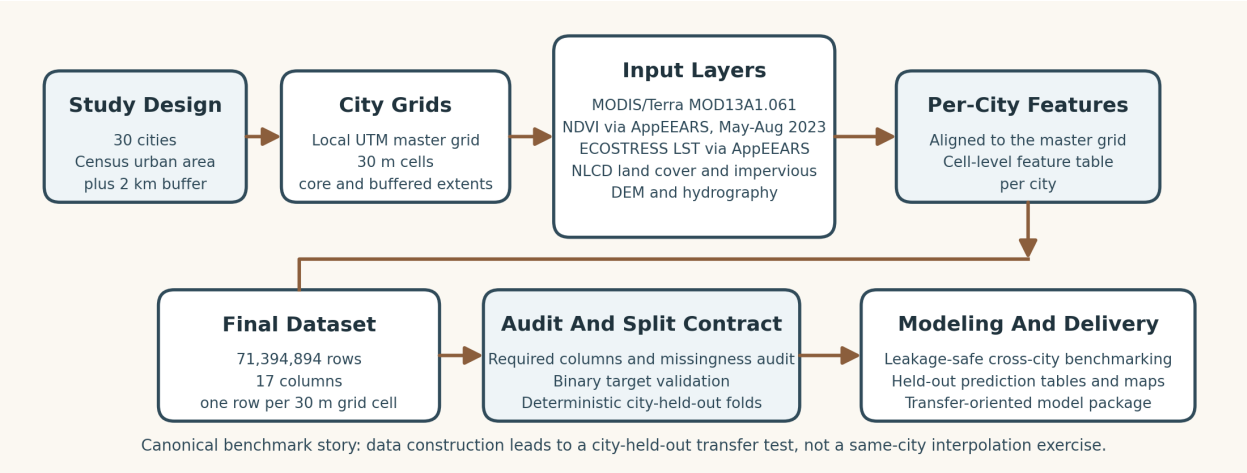
Metric	Mean delta	Median delta	SD delta	Min delta	Max delta	RF wins	Logistic wins	Ties
City PR AUC	-0.0023	-0.0136	0.0602	-0.0743	0.1945	9	21	0
City re-call@top10	+0.0106	-0.0150	0.0901	-0.0700	0.3420	9	21	0

Figure 1. Study City Locations



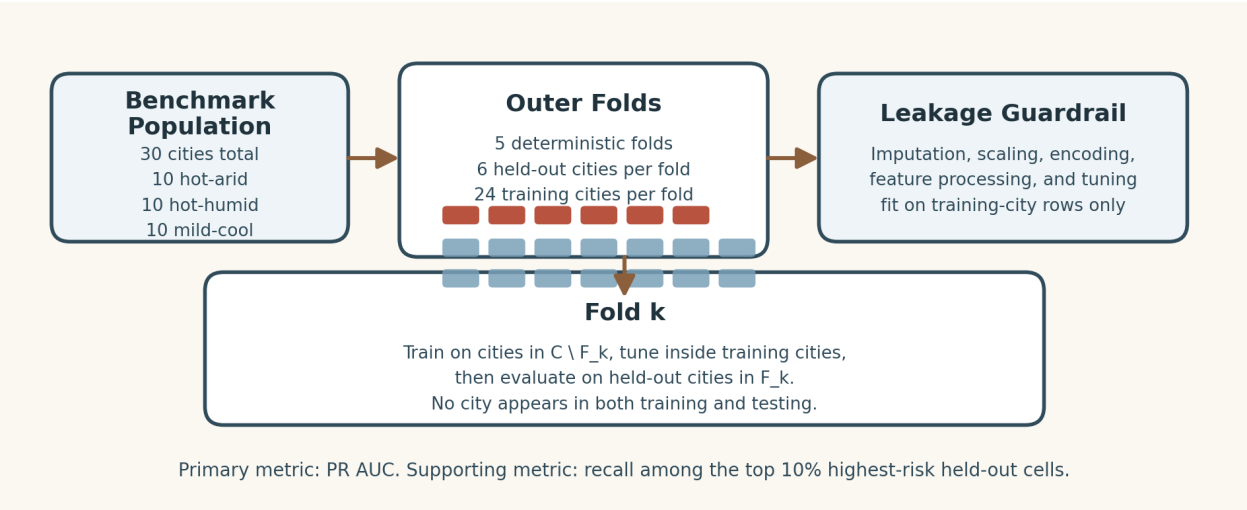
The 30 benchmark cities span western, southern, and northern U.S. regions and are colored by broad climate group. Appendix Table A4 lists full city names, climate groups, row counts, and fold assignments.

Figure 2. Dataset Construction Workflow



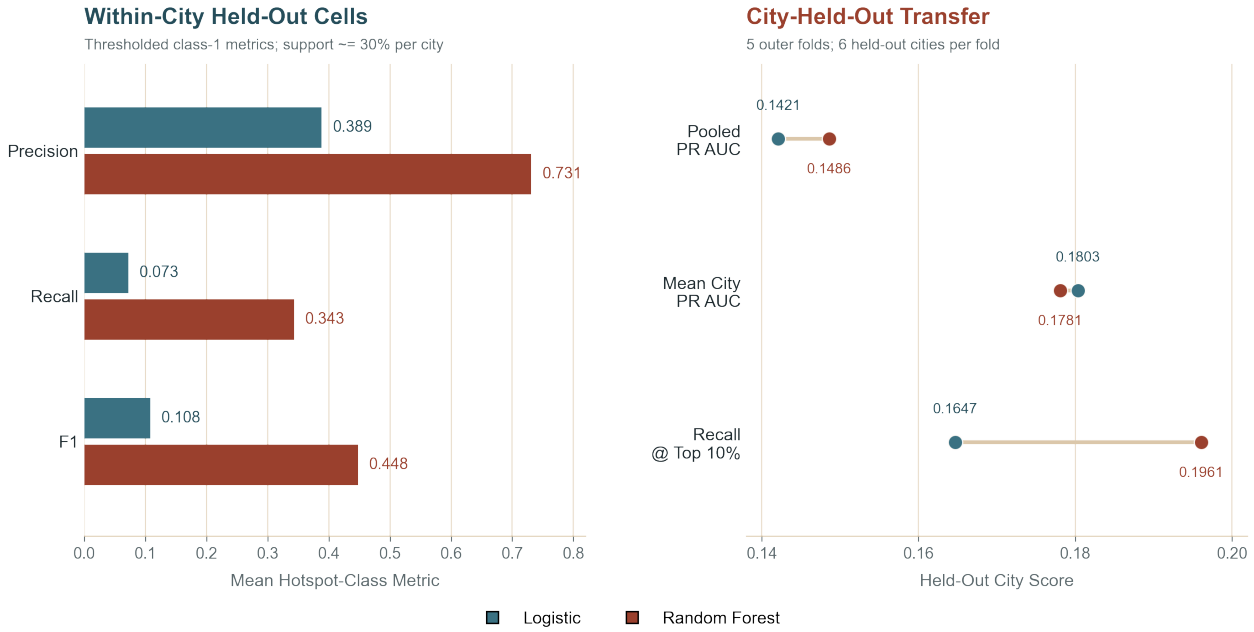
Buffered Census urban-area study regions and local 30 m grids feed aligned per-city feature assembly, including MODIS/Terra MOD13A1.061 NDVI and ECOSTRESS LST acquired through AppEEARS for May-August 2023, followed by final dataset generation, audit, city-held-out folds, and model evaluation.

Figure 3. City-Held-Out Evaluation Design



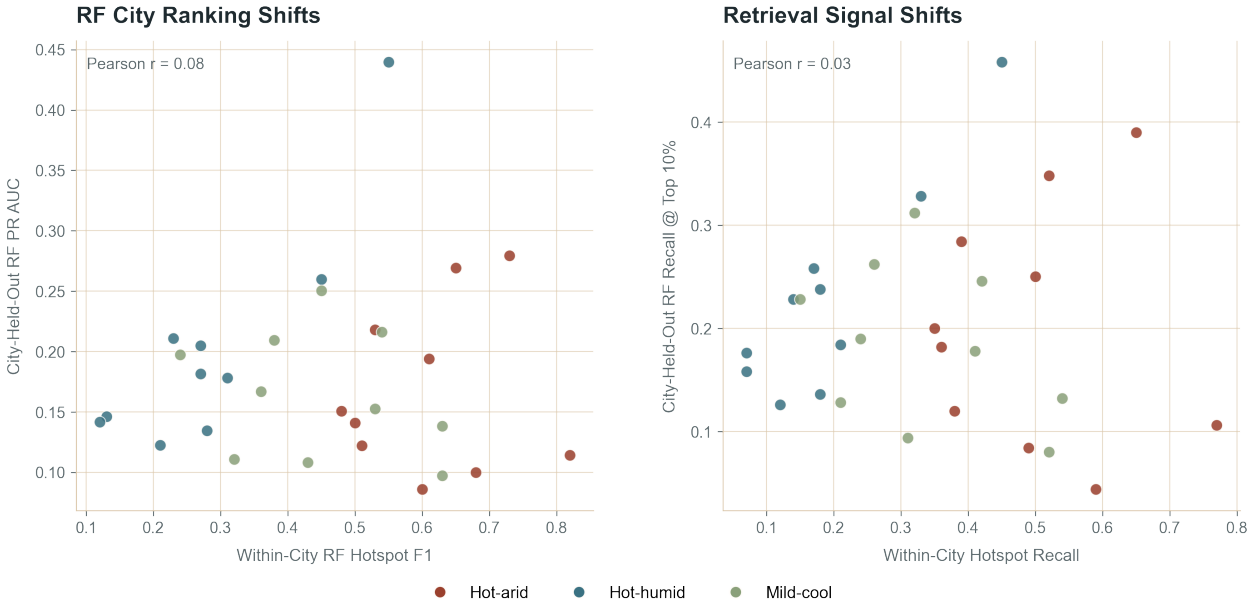
Each outer fold holds out six complete cities and trains on the remaining 24 cities. All preprocessing and tuning are fit using training-city rows only.

Figure 4. Within-City and City-Held-Out Results Side by Side



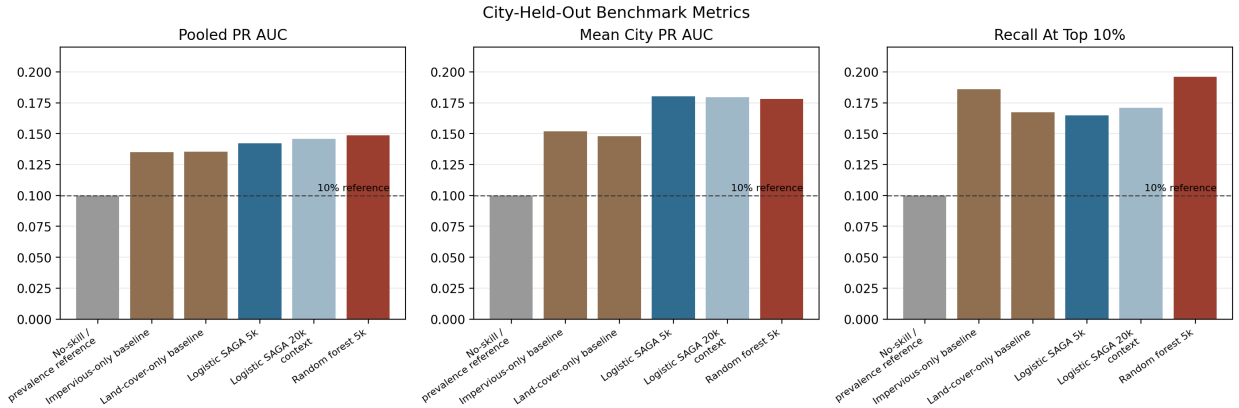
The left panel summarizes the verified within-city 70/30 held-out result using hotspot precision, recall, and F1. The right panel summarizes the retained city-held-out transfer benchmark using pooled PR AUC, mean city PR AUC, and recall at top 10%. The figure compares validation designs and model-ranking patterns, not direct metric magnitudes across the two panels.

Figure 5. City-Level Signal Shifts Across Evaluation Designs



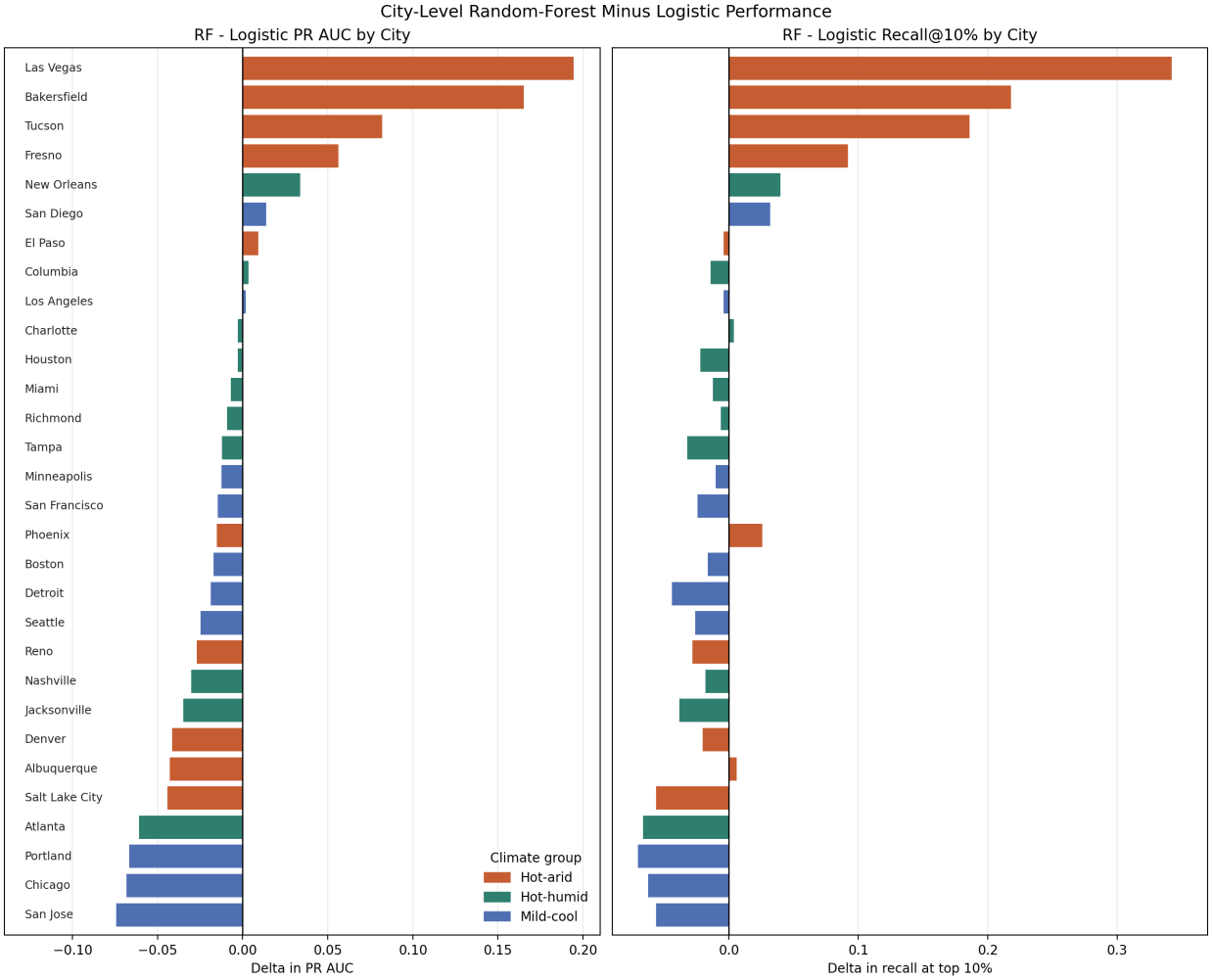
Each point is one city. The weak city-level correlations show that within-city random-forest success does not reliably predict city-held-out random-forest success. The left panel compares within-city RF hotspot F1 with transfer PR AUC; the right panel compares within-city RF hotspot recall with transfer recall at top 10%.

Figure 6. City-Held-Out Benchmark Metric Comparison



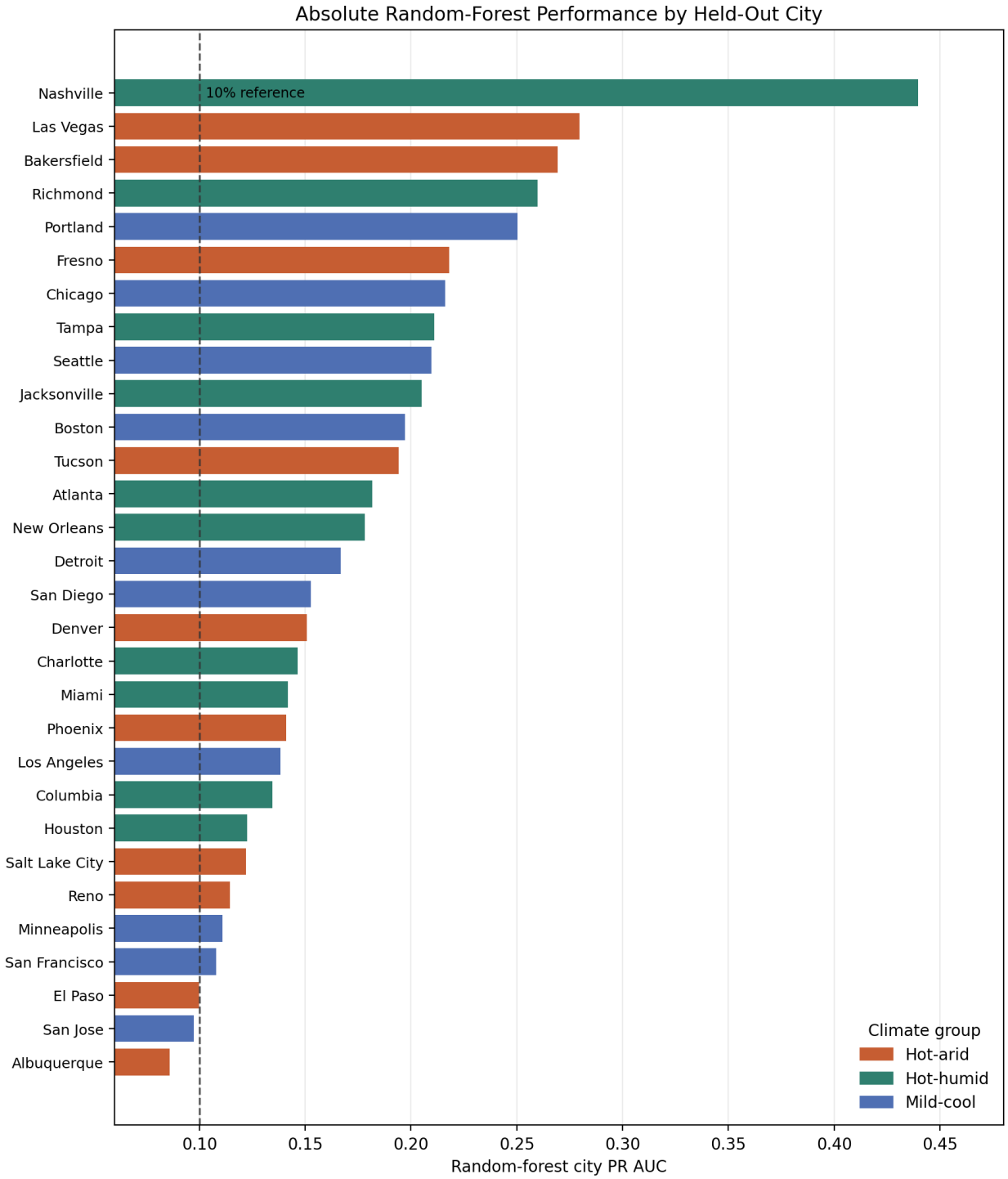
Sampled city-held-out benchmark metrics compare the no-skill reference, simple baselines, matched 5k logistic and random-forest models, and the 20k logistic context run. The dashed line marks the 10% prevalence reference. The learned models show modest gains over the strongest simple baselines rather than a large separation.

Figure 7. City-Level RF Minus Logistic Deltas



Horizontal bars show random-forest minus logistic deltas for each city, sorted by PR AUC delta and colored by climate group. In the aligned report narrative, this figure is a transfer-heterogeneity diagnostic, not the whole project conclusion. Random-forest gains are concentrated in Las Vegas, Bakersfield, Tucson, and Fresno, while San Jose, Chicago, Portland, and Atlanta are among the clearest logistic-favoring cities.

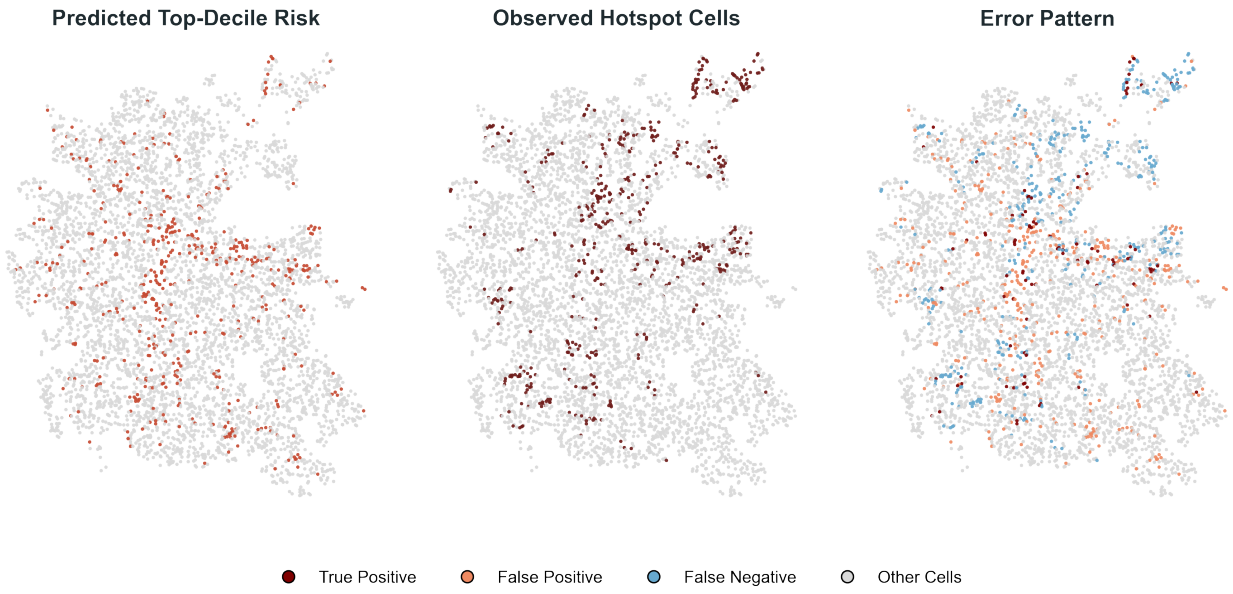
Figure 8. Absolute Random-Forest City PR AUC



Horizontal bars show retained random-forest PR AUC for each held-out city, sorted by absolute city PR AUC and colored by climate group. The dashed line marks the 0.10 reference implied by the sampled 10% hotspot prevalence. This absolute-performance view prevents interpreting Figure 7's RF-minus-logistic gains as a direct ranking of which cities

are easiest for the random forest to predict.

Figure 9. Held-Out Denver Map Example



Representative held-out Denver benchmark snapshot showing predicted top-decile risk, true hotspot cells, and categorical error types for the retained random-forest model. Denver is used as a representative hot-arid held-out example from the retained map outputs. False positives and false negatives remain spatially structured, suggesting partial built-environment signal but missed city-specific detail.

Appendix

Appendix Table A1. Final Dataset Columns

Column	Definition	Role in report	Used in headline model?
`city_id`	Integer city identifier used for joins and grouped cross-validation.	Grouping / metadata	No
`city_name`	Human-readable city name.	Metadata	No
`climate_group`	Broad climate grouping label for the city.	Predictor / stratifier	Yes
`cell_id`	Cell identifier within the city grid.	Cell metadata	No
`centroid_lon`	Cell centroid longitude in WGS84.	Mapping metadata	No
`centroid_lat`	Cell centroid latitude in WGS84.	Mapping metadata	No
`impervious_pct`	NLCD impervious percentage for the cell.	Predictor	Yes
`land_cover_class`	NLCD land-cover class code for the cell.	Predictor / water filter	Yes
`elevation_m`	DEM-derived elevation in meters.	Predictor	Yes
`dist_to_water_m`	Distance from the cell to the nearest hydro feature in meters.	Predictor	Yes
`ndvi_median_may_aug`	Median May-August NDVI derived from AppEEARS MODIS/Terra MOD13A1.061 inputs.	Predictor	Yes
`lst_median_may_aug`	Median May-August 2023 daytime land surface temperature derived from ECOSTRESS/AppEEARS inputs.	Outcome ingredient	No
`n_valid_ecostress_passes`	Number of valid ECOSTRESS observations contributing to the cell-level LST summary.	Quality filter / support field	No

Column	Definition	Role in report	Used in headline model?
`hotspot_10pct`	Binary indicator for whether the cell falls in the within-city top 10% of valid LST values.	Target	Target only
`tree_cover_proxy_pct_270m`	Share of nearby 30 m cells within an approximately 270 m neighborhood in NLCD forest classes 41/42/43.	Supplemental neighborhood-context feature	No
`vegetated_cover_proxy_pct_270m`	Share of nearby 30 m cells within an approximately 270 m neighborhood in selected NLCD vegetated classes.	Supplemental neighborhood-context feature	No
`impervious_pct_mean_270m`	Neighborhood mean NLCD impervious percentage within an approximately 270 m window.	Supplemental neighborhood-context feature	No

Appendix Table A2. Model Run Metadata

Model checkpoint	Preset	Rows per city	Outer folds	Inner CV splits	Candidate settings	Estimated inner fits	Scoring
Logistic SAGA 5k	full	5,000	0-4	4	20	400	average precision
Random forest 5k	targeted RF search	5,000	0-4	3	8	120	average precision

Metrics: Logistic SAGA 5k reached PR AUC 0.1421, mean city PR AUC 0.1803, and recall@top10 0.1647. Random forest 5k reached PR AUC 0.1486, mean city PR AUC 0.1781, and recall@top10 0.1961.

The logistic run tuned regularization strength and `l1_ratio` values corresponding to L2, L1, and elastic-net variants. The random-forest run tuned tree count, maximum depth, feature subsampling, and minimum leaf size. Selected hyperparameters varied by outer fold.

Appendix Table A3. Model and Baseline Specifications

Model / baseline	Predictors	Preprocessing	Tuning grid or rule	Scoring	Grouped CV?
No-skill / prevalence reference	None	None	Reference PR AUC and top-decile recall equal to the 10% target rate.	PR AUC and recall@top10	Reference only
Global-mean baseline	None	Training-city target mean	Predict the training-city hotspot prevalence for all held-out rows.	PR AUC and recall@top10	Outer city folds only
Climate-only baseline	`climate_group`	Training-city category means	Predict training-city hotspot prevalence by climate group.	PR AUC and recall@top10	Outer city folds only
Land-cover-only baseline	`land_cover_class`	Training-city category means	Predict training-city hotspot prevalence by land-cover class.	PR AUC and recall@top10	Outer city folds only
Impervious-only baseline	`impervious_pct`	Training-city decile bins	Predict training-city hotspot prevalence by imperviousness bin.	PR AUC and recall@top10	Outer city folds only
Logistic SAGA 5k	Six non-thermal predictors	Training-only imputation, numeric scaling, and categorical one-hot encoding inside sklearn Pipeline	`C` = 0.01, 0.1, 1.0, 10.0; `l1_ratio` = 0.0, 0.2, 0.5, 0.8, 1.0	Inner-CV average precision; held-out PR AUC and recall@top10	Yes, grouped outer folds and grouped inner CV

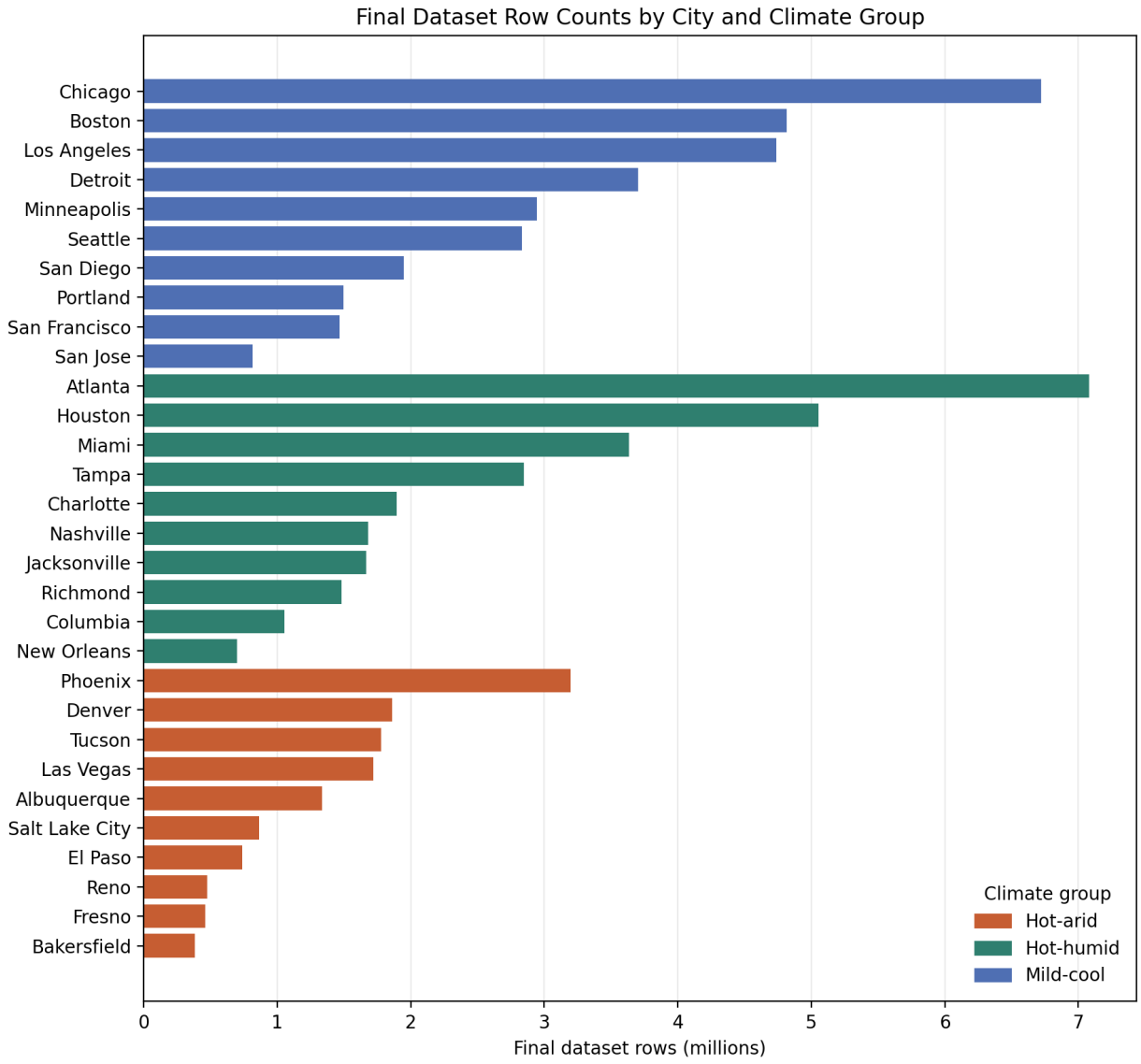
Model / baseline	Predictors	Preprocessing	Tuning grid or rule	Scoring	Grouped CV?
Random forest 5k	Same six non-thermal predictors	Training-only imputation and categorical one-hot encoding inside sklearn Pipeline	`n_estimators` = 200, 300; `max_depth` = 10, 20; `max_features` = sqrt; `min_samples_leaf` = 1, 5	Inner-CV average precision; held-out PR AUC and recall@top10	Yes, grouped outer folds and grouped inner CV

Appendix Table A4. City and Fold Composition

City ID	City	Climate group	Final rows	Hotspot count	Hotspot prevalence	Outer fold
1	Phoenix	Hot-arid	3,199,440	319,949	0.1000	2
2	Tucson	Hot-arid	1,779,906	177,991	0.1000	0
3	Las Vegas	Hot-arid	1,718,669	171,867	0.1000	3
4	Albuquerque	Hot-arid	1,336,755	133,676	0.1000	4
5	El Paso	Hot-arid	738,527	73,853	0.1000	4
6	Denver	Hot-arid	1,859,393	185,943	0.1000	1
7	Salt Lake City	Hot-arid	863,557	86,356	0.1000	1
8	Fresno	Hot-arid	459,104	45,912	0.1000	1
9	Bakersfield	Hot-arid	382,964	38,297	0.1000	0
10	Reno	Hot-arid	475,828	47,583	0.1000	2
11	Houston	Hot-humid	5,054,661	505,468	0.1000	2
12	Columbia	Hot-humid	1,055,916	105,626	0.1000	2
13	Richmond	Hot-humid	1,481,846	148,185	0.1000	0
14	New Orleans	Hot-humid	700,063	70,008	0.1000	3
15	Tampa	Hot-humid	2,847,118	284,712	0.1000	0
16	Miami	Hot-humid	3,635,068	363,510	0.1000	3
17	Jacksonville	Hot-humid	1,664,542	166,458	0.1000	2
18	Atlanta	Hot-humid	7,081,699	708,171	0.1000	0
19	Charlotte	Hot-humid	1,896,996	189,703	0.1000	3

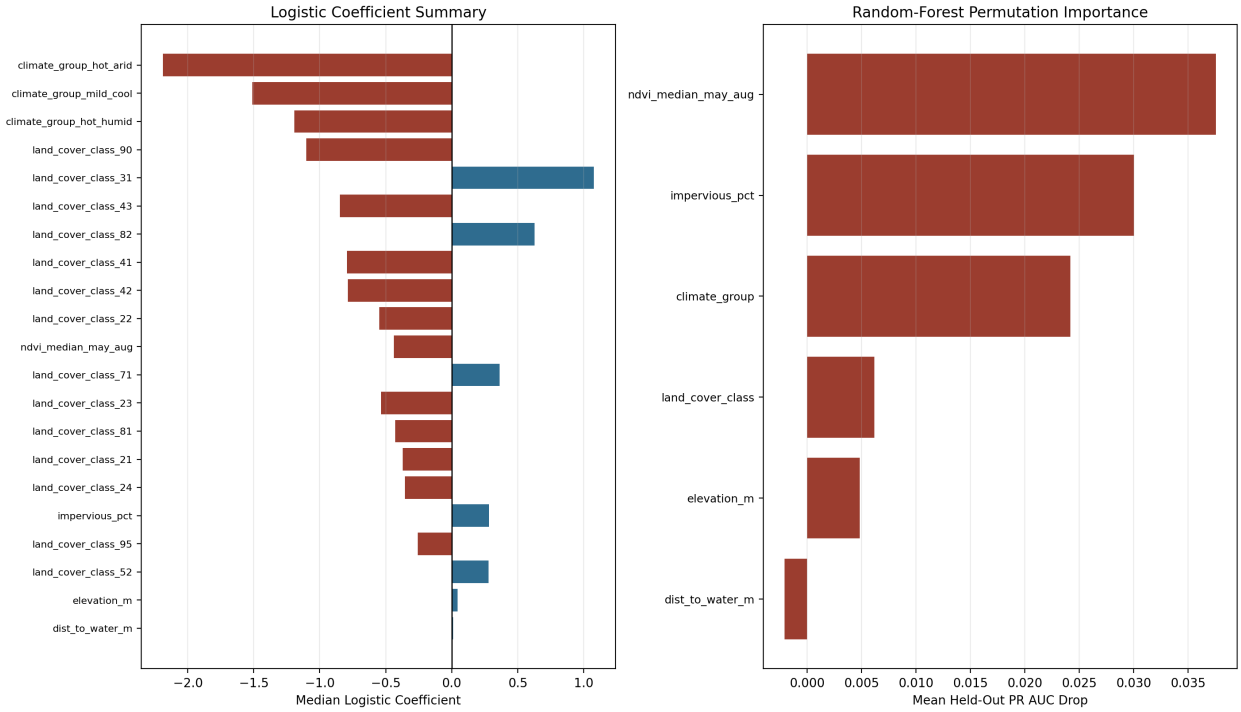
City ID	City	Climate group	Final rows	Hotspot count	Hotspot prevalence	Outer fold
20	Nashville	Hot-humid	1,680,248	168,025	0.1000	4
21	Seattle	Mild-cool	2,831,875	283,189	0.1000	2
22	Portland	Mild-cool	1,496,116	149,618	0.1000	1
23	San Francisco	Mild-cool	1,466,276	146,628	0.1000	3
24	San Jose	Mild-cool	817,627	81,764	0.1000	0
25	Los Angeles	Mild-cool	4,736,063	473,607	0.1000	4
26	San Diego	Mild-cool	1,948,679	194,869	0.1000	4
27	Chicago	Mild-cool	6,722,963	672,306	0.1000	1
28	Minneapolis	Mild-cool	2,946,162	294,619	0.1000	1
29	Detroit	Mild-cool	3,702,849	370,291	0.1000	4
30	Boston	Mild-cool	4,813,984	481,404	0.1000	3

Appendix Figure A1. Final Dataset Row Counts by City and Climate Group



Final row counts vary substantially by city because buffered study-area extents differ.

Appendix Figure A2. Supplemental Feature-Importance Summary



Permutation importance and coefficient summaries are included as predictive diagnostics only. They are not causal estimates of how changing an urban feature would change LST. NLCD land-cover categories are encoded categorical levels, so logistic coefficient signs depend on the reference category.

Reproducibility Notes

Report-facing tables and generated figures are regenerated with:

```
C:\Users\golde\.venvs\STAT5630_FinalProject_DataProcessing\Scripts\python.exe -m src.run
```

Canonical dataset and modeling-prep artifacts:

- `data_processed/final/final_dataset.parquet`
- `data_processed/final/final_dataset.csv`
- `data_processed/modeling/final_dataset_audit.md`
- `data_processed/modeling/final_dataset_audit_summary.json`
- `data_processed/modeling/final_dataset_city_summary.csv`
- `data_processed/modeling/city_outer_folds.parquet`
- `data_processed/modeling/city_outer_folds.csv`

Report table and figure artifacts:

- docs/report/tables/data_sources_variables.csv
- docs/report/tables/final_dataset_by_climate_group.csv
- docs/report/tables/benchmark_metrics.csv
- docs/report/tables/rf_vs_logistic_by_climate.csv
- docs/report/tables/rf_vs_logistic_by_fold.csv
- docs/report/tables/rf_vs_logistic_city_paired_summary.csv
- docs/report/tables/final_dataset_columns.csv
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- docs/report/tables/city_fold_composition.csv
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- docs/report/figures/benchmark_metrics.png
- docs/report/figures/city_metric_deltas.png
- docs/report/figures/city_rf_pr_auc.png
- docs/report/figures/heldout_denver_map_focus.png
- docs/report/figures/feature_importance_ranked_summary.png
- docs/report/figures/final_dataset_city_row_counts.png

The PDF render uses PNG figure files so the report does not depend on local SVG conversion support.

Benchmark source artifacts:

- outputs/modeling/reporting/cross_city_benchmark_report.md
- outputs/modeling/reporting/tables/cross_city_benchmark_report_benchmark_table.csv
- outputs/modeling/reporting/tables/cross_city_benchmark_report_city_error_by_climate
- outputs/modeling/reporting/tables/cross_city_benchmark_report_city_error_comparison
- outputs/modeling/logistic_saga/full_allfolds_s5000_sampled-full-allfolds_2026-04-07
- outputs/modeling/logistic_saga/full_allfolds_s20000_samplecurve-20k_2026-04-08_0211
- retained random-forest 5k run directory listed in Appendix Table A2 metadata

Supplemental interpretation and map artifacts, if used, should remain explicitly supplemental:

- outputs/modeling/supplemental/feature_importance/feature_importance_summary.md
- outputs/modeling/reporting/heldout_city_maps/heldout_city_maps.md